XIANG LI

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EDUCATION

Peking University, Beijing, China *Sep. 2018 – Jul. 2023*
Ph.D. in Statistics, School of Mathematical Sciences
Advisor: Prof. Zhihua Zhang

Peking University, Beijing, China *Sep. 2014 – Jul. 2018*
B.S. in Statistics, School of Mathematical Sciences
B.A. in Economics, National School of Development

EXPERIENCE

University of Pennsylvania, Philadelphia, USA *Aug. 2023 – Present*
Postdoctoral Researcher, Department of Statistics and Data Science
Working with Prof. Weijie Su and Prof. Qi Long

University of Toronto, Toronto, Canada *Feb. 2022 – Feb. 2023*
Visiting Ph.D. Student, Department of Statistical Sciences
Working with Prof. Qiang Sun

Face++ (Megvii), Beijing, China *Jun. 2018 – Nov. 2018*
Research Intern, Algorithm Group
Mentor: Dr. Shuchang Zhou

ToSimple, Beijing, China *Mar. 2018 – May. 2018*
Research Intern, Algorithm Group
Mentor: Dr. Naiyan Wang

SELECTED AWARDS

- IMS New Researcher Travel Award *2025*
- Outstanding PhD Graduates, Peking University *2023*
- President Scholarship for Excellent Ph.D Student, Peking University *2018, 2020-2021*
- Outstanding reviewer in NeurIPS, top 8% *2021*
- National Scholarship, China *2017, 2019*
- Travel Award *NeurIPS 2019, AAAI 2020*
- Outstanding Graduates, School of Mathematical Sciences, Peking University *2018*
- First prize in National Undergraduate Mathematical Modeling Contest, China *2016*

RESEARCH INTERESTS

My research lies at the intersection of statistical machine learning, optimization, and trustworthy AI, aiming to bridge theoretical foundations with practical impact. I am currently interested in developing principled and scalable methods for modern AI systems, especially large language models (LLMs).

My current and past research spans the following directions:

- **Statistical Foundations of Generative AI (Current)**: Develop statistical and algorithmic principles for generative models, investigating fundamental issues such as model evaluation, uncertainty quantification, and model collapse. The long-term goal is to provide theoretically grounded tools to better understand and improve model performance.
- **Safety in Generative Models (Current)**: Address AI safety challenges by designing watermarking and detection methods for LLMs that remain reliable under human edits and adversarial modifications. The aim is to establish principled mechanisms for tracing AI outputs and ensuring responsible deployment.
- **Robustness in Learning Systems (Ph.D.)**: Developed algorithms that remain effective under heavy-tailed noise, corruption, and heterogeneity, combining provable guarantees with practical efficiency.
- **Uncertainty Quantification in Online Learning (Ph.D.)**: Developed statistical methods for streaming and sequential settings, including confidence intervals for reinforcement learning, federated learning, and adaptive decision-making. These methods are useful for ensuring reliable decisions in dynamic environments.
- **Optimization for Decentralized and Federated Learning (Ph.D.)**: Studied optimization problems in distributed systems, with emphasis on communication-efficient algorithms and convergence behavior under data heterogeneity.

PUBLICATIONS

* indicates equal contribution; ** indicates alphabetical order.

Journal papers

[J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Annals of Statistics, 53 (1): 322-351, Feb. 2025

[J2] **Robust Detection of Watermarks for Large Language Models Under Human Edits** [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
To appear in *Journal of the Royal Statistical Society Series B*

[J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su
Major revision at *Journal of the American Statistical Association.*

[J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [pdf] [arxiv]
Yangxinyu Xie, **Xiang Li**, Tanwi Mallick, Weijie J. Su, Ruixun Zhang
Journal of the American Statistical Association, 1–11, 2025

[J5] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [arxiv]
Xiang Li*, Jiadong Liang*, Xinyun Chen, Zhihua Zhang
Minor revision at *Operation Research*

[J6] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [pdf]
Yuze Han**, **Xiang Li****, Shiyun Lin**, Zhihua Zhang**
Journal of Machine Learning Research, 25 (380): 1-88, Dec. 2024

[J7] **Variance-aware Decision Making with Linear Function Approximation with Heavy-tailed Rewards** [pdf] [arxiv]
Xiang Li, Qiang Sun
Transactions on Machine Learning Research, Apr. 2024

[J8] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [pdf] [arxiv]
Xiao Guo*, **Xiang Li***, Xiangyu Chang, Shusen Wang, Zhihua Zhang
Machine Learning, 113 (11): 8427-8458, Sep. 2024

Conference papers

[C1] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [pdf]
Xiang Li, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C2] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [pdf]
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun
Conference on Uncertainty in Artificial Intelligence (UAI) 2025

[C3] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [pdf]
Hengchao Chen, **Xiang Li**, Qiang Sun
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C4] **Statistical Estimation and Online Inference via Local SGD** [pdf]
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang
Conference on Learning Theory (COLT) 2022

[C5] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [pdf]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022, Spotlights

[C6] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [pdf]
Shiyun Lin*, Yuze Han*, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022

[C7] **Communication-Efficient Distributed SVD via Local Power Iterations** [pdf]
Xiang Li, Shusen Wang, Kun Chen, Zhihua Zhang
International Conference on Machine Learning (ICML) 2021

[C8] **Finding the Near Optimal Policy via Reductive Regularization in MDPs** [pdf]
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang
Workshop on Reinforcement Learning Theory, ICML 2021

[C9] **On the Convergence of FedAvg on Non-IID Data** [pdf]
Xiang Li*, Kaixuan Huang*, Wenhao Yang*, Shusen Wang, Zhihua Zhang
International Conference on Learning Representations (ICLR) 2020, Oral presentation

[C10] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [pdf]
Xiang Li, Shusen Wang, Zhihua Zhang
AAAI Conference on Artificial Intelligence (AAAI) 2020

[C11] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [pdf]
Wenhao Yang*, **Xiang Li***, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2019

Submitted

[S1] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [arxiv]
Xiang Li, Jiayi Xin, Qi Long, Weijie J. Su

[S2] **Uncertainty Quantification of Data Shapley via Statistical Inference** [pdf]
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang

[S3] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han**, **Xiang Li****, Zhihua Zhang**

[S4] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang

[S5] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [arxiv]
Xiang Li, Jiadong Liang, Zhihua Zhang

[S6] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [arxiv]
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma

Technical reports

[T1] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [arxiv]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang

[T2] **Communication-Efficient Local Decentralized SGD Methods** [arxiv]
Xiang Li, Wenhao Yang, Shusen Wang, Zhihua Zhang

PRESENTATIONS

- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
  - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
  - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.
- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*

- Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.

- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

ACADEMIC SERVICE

- Conference review: NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.
- Journal review: Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Journal of the American Statistical Association (JASA), Annals of Applied Probability (AOAP), Operational Research (OR), IEEE Transactions on Automatic Control, IEEE Open Journal of Signal Processing, IEEE Journal on Selected Areas in Communications, Transactions on Parallel and Distributed Systems, World Wide Web (WWW).
- Event organization:
  - Aug. 2024, Session Chair for "Theoretical and Algorithmic Developments in Federated Learning" at the Modeling and Optimization: Theory and Applications (MOPTA) conference.

TEACHING EXPERIENCES

- Lecturer, Short course at the 2025 ICSA Applied Statistics Symposium, Summer 2025 [slides]
- Teaching Assistant, *Linear Algebra*, Spring 2021
- Teaching Assistant, *Reinforcement Learning: Theory and Algorithms*, Fall 2019